LMS Live PostgreSQL Entity-Relationship Diagram

Generated directly from live PostgreSQL metadata • PK, FK, unique constraints and column types

Identity, Faculty & Department

faculties
- [PK] faculty_id: INTEGER
- [UQ] name: VARCHAR(255)
- [UQ] code: VARCHAR(50)

departments
- [PK] department_id: INTEGER
- [FK] [UQ] faculty_id: INTEGER
- [UQ] name: VARCHAR(255)
- [UQ] code: VARCHAR(50)

users
- [PK] user_id: INTEGER
- username: VARCHAR(50)
- email: VARCHAR(255)
- password_hash: VARCHAR(255)
- account_status: VARCHAR(50)
- created_at: TIMESTAMP
- [FK] department_id: INTEGER

roles
- [PK] role_id: INTEGER
- name: VARCHAR(50)

user_roles
- [PK] [FK] user_id: INTEGER
- [PK] [FK] role_id: INTEGER
- assigned_at: TIMESTAMP
- [FK] assigned_by: INTEGER

lecturer_approval_events
- [PK] approval_id: INTEGER
- [FK] lecturer_user_id: INTEGER
- [FK] administrator_user_id: INTEGER
- decision: VARCHAR(50)
- decided_at: TIMESTAMP

Courses & Materials

courses
- [PK] course_id: INTEGER
- title: VARCHAR(255)
- [FK] owner_user_id: INTEGER
- ai_enabled: BOOLEAN
- [FK] department_id: INTEGER

enrolments
- [PK] [FK] course_id: INTEGER
- [PK] [FK] user_id: INTEGER
- status: VARCHAR(50)
- enrolled_at: TIMESTAMP

concepts
- [PK] concept_id: INTEGER
- [FK] [UQ] course_id: INTEGER
- [UQ] name: VARCHAR(255)

concept_prerequisites
- [PK] [FK] concept_id: INTEGER
- [PK] [FK] prerequisite_concept_id: INTEGER

materials
- [PK] material_id: INTEGER
- [FK] course_id: INTEGER
- [FK] uploaded_by: INTEGER
- file_path: VARCHAR(512)
- processing_status: VARCHAR(50)
- uploaded_at: TIMESTAMP

material_sections
- [PK] section_id: INTEGER
- [FK] [UQ] material_id: INTEGER
- [UQ] section_order: INTEGER
- text: TEXT
- source_location_type: VARCHAR(50)
- source_location_value: VARCHAR(100)

material_section_concepts
- [PK] [FK] section_id: INTEGER
- [PK] [FK] concept_id: INTEGER

Assessment & Grading

assignments
- [PK] assignment_id: INTEGER
- [FK] course_id: INTEGER
- title: VARCHAR(255)
- due_at: TIMESTAMP
- lifecycle_status: VARCHAR(50)
- results_released_at: TIMESTAMP
- [FK] results_released_by: INTEGER
- cmb_enabled: BOOLEAN

assignment_questions
- [PK] question_id: INTEGER
- [FK] [UQ] assignment_id: INTEGER
- prompt: TEXT
- max_score: NUMERIC(5, 2)
- [UQ] position: INTEGER

question_concepts
- [PK] [FK] question_id: INTEGER
- [PK] [FK] concept_id: INTEGER

submissions
- [PK] submission_id: INTEGER
- [FK] [UQ] assignment_id: INTEGER
- [FK] [UQ] user_id: INTEGER
- submitted_at: TIMESTAMP
- work_reference: VARCHAR(512)
- marking_status: VARCHAR(50)

marks
- [PK] [FK] submission_id: INTEGER
- [PK] [FK] question_id: INTEGER
- score: NUMERIC(5, 2)
- lecturer_comment: TEXT
- [FK] graded_by: INTEGER
- graded_at: TIMESTAMP

Concept-to-Mastery Bridge

cmb_items
- [PK] item_id: INTEGER
- [FK] course_id: INTEGER
- item_type: VARCHAR(50)
- prompt: TEXT
- review_status: VARCHAR(50)
- [FK] approved_by: INTEGER

cmb_item_concepts
- [PK] [FK] item_id: INTEGER
- [PK] [FK] concept_id: INTEGER

diagnostic_sessions
- [PK] session_id: INTEGER
- [FK] submission_id: INTEGER
- started_at: TIMESTAMP
- completed_at: TIMESTAMP

responses
- [PK] response_id: INTEGER
- [FK] session_id: INTEGER
- [FK] item_id: INTEGER
- answer: TEXT
- confidence_choice: VARCHAR(50)
- response_time: INTEGER

candidate_explanations
- [PK] explanation_id: INTEGER
- [FK] session_id: INTEGER
- [FK] concept_id: INTEGER
- label: VARCHAR(255)
- confidence: DOUBLE PRECISION

interventions
- [PK] intervention_id: INTEGER
- [FK] explanation_id: INTEGER

recommended_resources
- [PK] [FK] intervention_id: INTEGER
- [PK] [FK] section_id: INTEGER
- reason: TEXT

followup_results
- [PK] followup_id: INTEGER
- [FK] intervention_id: INTEGER
- [FK] item_id: INTEGER
- correct: BOOLEAN
- answered_at: TIMESTAMP

Audit & AI Gateway

access_logs
- [PK] log_id: INTEGER
- [FK] user_id: INTEGER
- action: VARCHAR(100)
- target_type: VARCHAR(100)
- target_id: VARCHAR(100)
- occurred_at: TIMESTAMP

ai_gateway_requests
- [PK] request_id: INTEGER
- [FK] user_id: INTEGER
- [FK] course_id: INTEGER
- purpose: VARCHAR(100)
- provider: VARCHAR(100)
- status: VARCHAR(50)
- created_at: TIMESTAMP

ai_gateway_request_sources
- [PK] [FK] request_id: INTEGER
- [PK] [FK] section_id: INTEGER

Legend: [PK] primary key • [FK] foreign key • [UQ] unique constraint • arrows point to referenced tables